Standard lithium batteries contain a metal oxide and a graphite electrode, with a shapeless polymer acting as the electrolyte. But a new solid-state polymer by St Andrews University, Scotland, promises greater densities and improved battery designs. The solid-state polymer serves as both the electrodes and the electrolyte, leading to greater densities and thinner battery designs. Under test conditions, solid-state polymers have shown to offer greater conductivity than amorphous polymers.

Scientists at the US Department of Energy have developed a new metal alloy that will enhance the life span of batteries. The alloy, made of nickel, tin and lanthanum, contains no toxins or the expensive cobalt and it will be cheaper and environment-friendly.

A battery of changes

The problem with the batteries of today is that they work on principles of electrochemistry that date back to the 18th century. The technology curve for them has plateaued out and even the best batteries have little chance of enhancing the power-to-weight ratio. Conventional batteries also contain toxic ingredients such as mercury and cadmium.

Fuel cells, which use hydrogen as a source, have a lot of inherent advantages. They produce only carbon dioxide, water and heat as waste products. Methanol and hydrogen-based fuel cells can provide 20 hours of talk time on a cell phone compared to the maximum airtime of four hours offered by lithium-based counterparts. Instead of plugging in your cell phone overnight, or swapping batteries, you need to simply snap in a new methanol cartridge.

Prototype of a Methanol fuel cell powered mobile

Fuel cells are like power plants, and the technology has larger applications such as powering buses and cars. Thanks to miniaturisation, the industry is gearing up to make fuel cells small enough for consumer electronics. Electronics giants such as Motorola, Samsung and NEC are already set on getting a piece of the action in this high-stakes industry. The competitors are betting on different designs, but methanol-based fuel cells promise applications that we are likely to see very soon. These fuel cells will sit at the back of battery compartments, and recharge the battery up to a month, eliminating the need for the battery's nightly snooze next to a wall socket.

Another long-term goal is to create fuel cells that store hydrogen and take oxygen from the atmosphere to create power. Many research institutes are simultaneously working on using pipe-like carbon molecules that have the ability to store and release hydrogen. Using Nanotechnology, researchers have created nano canisters filled with hydrogen encased in micron-sized carbon tubes. There have been credible results to date at the MIT and the Chinese Academy of Sciences, where researchers have reportedly made carbon nanotubes that can store 4.2 per cent of their weight in hydrogen.

Electronics giants too are knee-deep in experiments with carbon molecules. Japan-based NEC Corporation, together with two Japanese research institutes, has developed a fuel cell for use in mobile devices that could enhance your notebook's battery life to around three days. The new fuel cell uses a type of nanotube called a nanohorn, discovered by the team three years ago. They are called so because of their horn-like shape, which lets them have a larger surface area, making them better suited for use as fuel cell electrodes. The new fuel cell has about 10 times the energy capacity of a similar-sized high-density lithium battery, such as those currently used in notebook computers or portable electronic devices. NEC is hoping to commercialise this technology sometime between 2003 and 2005.

Aluminium power

Aluminum-Power Inc, Canada has developed an aluminium-air fuel cell that can revolutionise the mobile electronics and automotive industries. A breakthrough technology that offers high, sustained power over extended periods, it's an environmentally friendly approach to powering emerging technologies and improving old ones.

While most other energy cell technologies

Zinc + Air = Power

The zinc-air battery has arisen as a new choice for handheld electronics, because it can outrun conventional alkaline batteries such as the Energiser by three times, while being more compact than its alkaline counterparts. Zinc-air cells work in the same way as conventional batteries do in that they generate electrical power from chemical reactions. But instead of packing the necessary ingredients into the cell, zinc-air batteries borrow one of their main components, oxygen, from the air outside.

Using a reactant from the air saves on space, reducing the size and weight of the battery. Zinc-air cells contain no toxic compounds and are neither highly reactive nor flammable. In fact, they can be recycled, safely disposed of, or in some cases, recharged with new zinc. A US-based company called Electric Fuel is developing zinc-air battery technology for automobiles. Instead of polluting exhaust fumes, cars could one day be breathing out air.